Final: Project Analysis/ Member Contributions

Team Name: Deep Learners

Teammates Involved:

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For the final project, the team is working on the Amazon Sales dataset from their mid-term project. In the midterm, we worked on data preparation by handling missing values, removing unnecessary columns, creating new features, and encoding categorical variables. In the final part of the project, we use clean data to build and test the machine learning models.

The work deep learners are doing in this project is the same process used in real businesses today. Companies like Amazon, Walmart, Target, and Best Buy are some businesses that use these machine learning models to:

1. Predict demand for products to understand how much inventory to order
2. Estimate future sales to help with budgeting and planning.
3. Set better prices, using models to suggest discounts or deal timing
4. Understand customers' behavior, e.g., which items are going to be best-sellers.
5. Improve recommendations by learning what shoppers are likely to buy.
6. Optimize marketing to predict which products need more promotion.

We exercised the same skills data scientists use to make decisions that save companies time, money, and resources by learning how to clean data, build models, compare those models, and combine them into ensembles.

John's Contribution

My contribution to this phase focused on setting up the modeling workflow and establishing baseline performance. I created the required 70/15/15 train/validation/test splits that the rest of the team relied on, then trained the Linear Regression baseline model and calculated MAE, MSE, and R^2 for both the validation and test sets. This gave the group a clear first point of comparison before moving on to tree-based and ensemble models.

A big part of the value in doing this step was seeing how the preprocessing pipeline, feature setup, and data split all tie together. Running the baseline helped me understand how even simple models reveal whether our data is behaving as expected, and why every project needs a baseline before experimenting with more complex algorithms. It also gave me more confidence in reading and interpreting regression metrics, which I didn't fully understand before this assignment.

Saima's Contribution

My work was to develop Decision Tree and Random Forest regression models to forecast which Amazon products would be purchased. The Decision Tree model constructs a large tree that splits the data based on various feature values. On the other hand, a Random Forest creates many trees and then averages their results, which helps stabilize the predictions and enhances their accuracy. The visual outputs produced in this regard include an important features plot; an actual vs predicted scatter plot, a residual plot, and performance metrics. The critical feature plot shows how essential characteristics such as rating, review, and discounts are in driving sales. Random Forest gives more stable rankings regarding feature importance. The actual versus predicted scatter plots depict how close our model predictions are to real prices. Points falling closer to the diagonal line show better accuracy, and Random Forest generally exhibits tighter clustering around this line compared to Decision Tree. Residual plots show how far predictions are from actual values. It helps us see if the model is making any consistent mistakes. Random Forest usually has residuals that are more randomly scattered around zero, which means it makes fewer systematic errors and tends to perform better than the Decision Tree model.

These visualizations are essential for real-world e-commerce decisions. For instance, the feature-importance charts show which product details matter most. For example, the businesses will improve their product quality and will gather more customer feedback if the driving force behind sales is ratings and reviews, and not lower prices. The actual vs predicted graphs also show how reliable the model is. When the points are close together, the predictions are more reliable, which is helpful for tasks like setting prices or managing inventory.

Misty's Richardson

My portion was to implement Gradient Boosting and KNN regressor with hyperparameter optimization and calculate their MAE, MSE, and R^2 metrics. To complete the random forest model, I implemented gradient boosting and the KNN regressor with a hyperparameter optimizer and evaluated performance using the same validation framework as in the midterm. In each model, I used the GridSearchCV to search over the predefined hyperparameter grid. During this, I looked at the number of estimators, the learning rate, and the tree depth for gradient boosting, the number of neighbors, and the weighting scheme for KNN, using cross-validation and neg_mean_squared_error as the scoring metric. After reviewing everything, I selected the hyperparameters I thought were best and fit them to the optimized model on the training data, then generated predictions on the validation set. When I assessed and compared the model performance, I calculated the MAE, MSE, and R^2 score for both the gradient boosting and KNN. Doing this allowed us, as a team, to assess the accuracy and goodness of fit of the decision tree and random forest models previously used.

Joseph's Contribution

I was tasked with implementing three ensemble methods: an average ensemble, a voting ensemble, and a Bayesian Ridge model as a Bayesian-style ensemble. The average ensemble combined the predictions of the top three models, Random Forest, Gradient Boosting, and K-Nearest Neighbors. In the voting ensemble, base estimators included Linear Regression, Random Forest, and Gradient Boosting. An overall vote determined the final prediction. The Bayesian Ridge model adopted a probabilistic framework, was initialized with priors, and was updated on the data to yield predictions that balance fit and uncertainty.

Predictions were made for both the validation and test sets, and MAE, MSE, and R^2 were calculated. The voting ensemble performed best on the validation data, with an R2 comparable to that of the top model among the individuals. The average ensemble was a little lower but still showed consistent performance across metrics. Notably, although the Bayesian Ridge model did not perform best on validation, it achieved the highest R^2 on the test set due to its probabilistic regularization and its handling of uncertainty.

This showed that ensemble methods can pay off, but different techniques will be most effective under various conditions. The voting ensemble had the best validation performance. At the same time, Bayesian Ridge generalized best to new data, providing strategies based on whether immediate accuracy or long-term stability is one's top priority.

Performance Summary

Validation Set — R^2 Comparison

Voting Ensemble	0.96639
Random Forest	0.96652 (best standalone
Average Ensemble	0.96335
Bayesian Ridge Ensemble	0.95831

Test Set — R^2 Comparison

Bayesian Ridge	0.94004 (best on test)
Voting Ensemble	0.93741
Average Ensemble	0.92421

Willie's Contribution

My responsibilities in this project were to evaluate and compare the performance of all five machine learning models. Keeping my focus specifically on Mean Absolute Error (MAE), Mean Squared Error (MSE), and R^2 Score (Coefficient of Determination). Using the previously generated predictions for the validation and test sets obtained by other team members, I computed the MAE, MSE, and R^2 for all five models (Linear Regression, Decision Tree, Random Forest, and Gradient Boosting). KNN was skipped in the code because it was distance-based rather than rule-based, and cannot compute feature importance values. I created a helper function to compute the three evaluation metrics in one place. This helped me easily reuse the same function by evaluating each model on its predictions for the validation and test sets. The summary tables clearly showed the results, and the comparison table allowed us to quickly identify which models had the lowest MAE/MSE and the highest R^2.

I combined all team members' contributions into one organized document and prepared our final presentation. I also authored the "Best Performance" report and the comprehensive "Full Approach" document, which describe the methodology we followed to achieve our outcomes.